7 C. Dispersal by animals –Zoochory (Example: *Achyranthes*)



Figure 7c: *Achyranthes*

Diagnostic Features

- Fruits are provided with hooks, spines, bristles, stiff hairs, etc. on their outer coat.
- The sharp pointed fruits of *Achyranthes* stick to the hairs of the animals and clothes and get carried away from one place to another.

Exercise 8: Ecological adaptations of plants found in hydrophytic, xerophytic, halophytic and epiphytic conditions.

Aim: To study plants found in different habitats and comment upon their adaptations.

Principle: The modifications in the structure of organisms to survive successfully in an environment are called adaptations of organisms. Observe different plants existing under various ecological habitats. The corresponding adaptations of plants and their interaction with the environment can be better understood.

Requirements: Fresh or preserved specimens of *Eichhornia*, *Opuntia*, *Avicennia* and *Vanda*.

8 A. Adaptations of Hydrophytes - *Eichhornia* (Water hyacinth)

Eichhornia is a free floating hydrophyte that grows in ponds, lakes and water bodies containing fresh water.

Diagnostic Features

- Root system is poorly developed.
- Root pockets are present.
- The petioles become swollen and spongy, providing buoyancy.
- Cortex is well developed with numerous air chambers. It helps in buoyancy and rapid gaseous exchange.
- Mechanical tissues are generally absent.

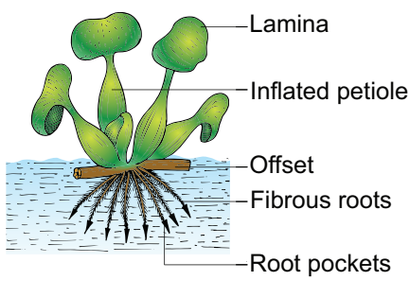


Figure 8a: Free floating hydrophyte – *Eichhornia*

8 B. Adaptations of Xerophytes - *Opuntia*

Opuntia is a succulent or drought resisting xerophyte, which grows wild in arid areas.

Diagnostic Features

- The stem is flattened, green, thick and fleshly called phylloclade
- Mucilage is present which helps to retain the water.
- Leaves are modified into spines

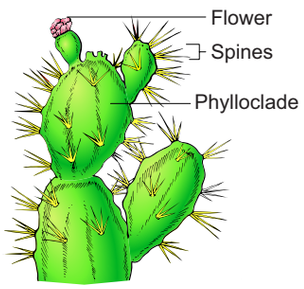


Figure 8a: Succulent xerophyte - *Opuntia*

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